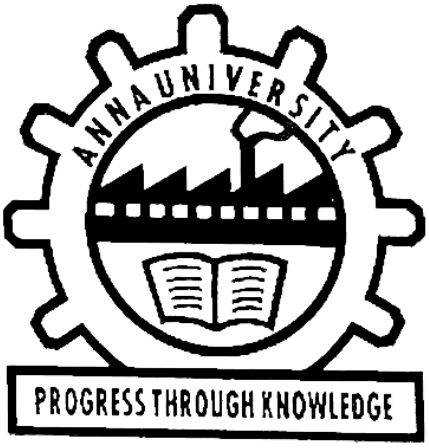
STATIC WEBSITE DEPLOYMENT USING AWS S3 CLOUD STORAGE

A SUMMER INTERNSHIP REPORT

Submitted by

K. SWATHI – 810023205037



DEPARTMENT OF INFORMATION TECHNOLOGY

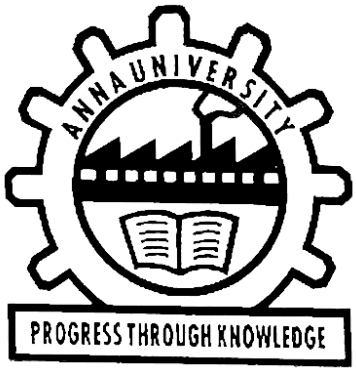
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BONAFIDE CERTIFICATE

This is to certify that this Summer Internship project report "**Static Website Deployment using AWS S3 Cloud Storage**" is the Bonafide work of **K. SWATHI (810023205037)** who has satisfactorily completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year **2026-2027**.

Faculty Incharge **Head of the Department**

Submitted for the practical examination held on _____

ABSTRACT

The **Static Website Deployment using AWS S3 Cloud Storage** project is a cloud computing internship project that demonstrates the complete process of hosting a website on the cloud. Cloud computing has changed the way software applications and digital content are hosted, managed and delivered by providing scalable computing resources on demand with minimal upfront investment. Among the services offered by **Amazon Web Services (AWS)**, **Amazon Simple Storage Service (Amazon S3)** stands out as a reliable, cost-effective and highly scalable object storage solution.

The project, titled "**CloudStore**", was developed during the one-month Cloud Computing Internship at **DLK Software Solutions, Chennai** (June – July 2026). The website was built using **HTML5 and CSS3** and represents a cloud-based file storage service. It features a navigation bar, a hero section with call-to-action buttons, a features section, a statistics panel, a pricing section, a step-by-step "How It Works" guide and an informative footer.

The deployment workflow involved **creating an AWS S3 bucket, disabling the public access block, uploading the HTML file, applying a JSON-based bucket policy for public read access and enabling static website hosting**. The website was then tested across multiple browsers and screen sizes to verify correct rendering and accessibility.

The project achieved its objectives: the website was deployed and verified as publicly accessible through the AWS S3 website endpoint, with **zero server management and near-zero operational cost within the AWS free tier**. For ease of access during evaluation, an identical build of the site is also mirrored on GitHub Pages. Overall, the project reinforces the practical utility of cloud hosting for lightweight static web content and lays the groundwork for more advanced cloud deployments.

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1. INTRODUCTION

The **Static Website Deployment using AWS S3 Cloud Storage** project is a cloud computing internship project carried out at DLK Software Solutions, Chennai. The world is undergoing a digital revolution driven by the rapid adoption of cloud computing technologies. Cloud computing has transformed the traditional IT landscape by shifting computing resources from physical on-premise infrastructure to virtualised, internet-accessible platforms managed by cloud providers. This shift has democratised access to enterprise-grade technology, allowing individuals, startups and small businesses to use the same powerful infrastructure as large corporations – at a fraction of the cost.

One of the most immediate and widely understood applications of cloud computing is **website hosting**. Traditionally, hosting a website required purchasing or leasing a physical or virtual server, configuring the server operating system, installing web server software, managing SSL certificates, and handling hardware failures and software updates. This process required technical expertise, time and significant financial investment.

Amazon S3 (Simple Storage Service) fundamentally changes this paradigm. It enables developers to host static websites – websites composed of HTML, CSS and JavaScript files – directly from a cloud storage bucket, without any server configuration whatsoever. The result is a website that is globally accessible, highly available, automatically scalable and virtually free for low-traffic use cases.

In this internship, a multi-section static website named **CloudStore** was designed using HTML5 and CSS3 and deployed on AWS S3 in the Asia Pacific (Mumbai) region. The report documents the complete workflow – from website design to bucket creation, file upload, bucket policy configuration, static website hosting and testing.

1.1 Objectives of the Internship

Specific Objectives

The main objective of the project is to **deploy a static website on the cloud using AWS S3** and to document the complete workflow as a learning resource. The objectives of the project are:

- To design and develop a multi-section, responsive static website using **HTML5 and CSS3**.
- To create and configure an **AWS S3 bucket** with static website hosting capability.
- To implement **JSON-based bucket policies** for controlled public access.
- To deploy the website and verify **global accessibility** through the S3 endpoint URL.
- To test the deployed website across **multiple browsers and screen sizes**.
- To understand the **AWS free tier** and the cost efficiency of cloud hosting.

- To document the complete workflow from design to deployment as a **learning resource**.
- To gain practical exposure to **cloud computing concepts** and AWS services in an industry environment.

1.2 Motivation

Small projects, internship demonstrations and prototype websites often do not require the complexity of full-stack server hosting. The challenge lies in finding a deployment method that is simple enough for students and beginners, yet robust and professional enough to serve as a real deployment solution. Traditional hosting options are either too expensive, too complex or too limited in scalability for demonstration and academic purposes.

This project addresses these challenges by deploying a static website using **AWS S3** – a platform that requires **no server management**, offers **built-in high availability** and falls entirely within the **AWS free tier** for small-scale use. It also provides hands-on experience in applying cloud computing concepts to a real deployment task.

1.3 Overview of Cloud Computing and AWS

Background. Cloud computing traces its roots to the 1960s, when John McCarthy proposed the idea of computation being delivered as a public utility. The modern cloud era began in 2006 when Amazon launched Amazon Web Services (AWS), offering scalable computing infrastructure to the public for the first time. This was followed by Google App Engine in 2008 and Microsoft Azure in 2010. According to Gartner, worldwide public cloud end-user spending reached over $597 billion in 2023 and is projected to exceed $1 trillion by 2027. AWS alone holds approximately 31% of the global cloud infrastructure market and serves millions of customers across 190 countries.

Definition. Cloud computing, as defined by NIST (2011), is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Essential characteristics of cloud computing

Characteristic	Description
On-Demand Self-Service	Users provision computing capabilities automatically without human interaction from the provider.
Broad Network Access	Resources are available over the network via standard platforms – phones, tablets, laptops.
Resource Pooling	Provider resources are pooled across multiple consumers using multi-tenancy.
Rapid Elasticity	Capabilities are elastically provisioned and released to scale with demand.
Measured Service	Resource use is controlled and optimised through metering – pay for what you use.

Cloud service models

Model	Full Name	Description
IaaS	Infrastructure as a Service	Virtualised hardware – compute, storage, networking. Ex: AWS EC2, S3.
PaaS	Platform as a Service	Platform for application development. Ex: AWS Elastic Beanstalk, Heroku.
SaaS	Software as a Service	Ready-to-use software delivered over the internet. Ex: Gmail, Dropbox.

This project uses AWS S3, which is an IaaS offering. The student manages the bucket configuration and files, while AWS manages the underlying storage infrastructure.

Cloud deployment models

Model	Description
Public Cloud	Third-party owned, shared infrastructure. Most cost-effective. Ex: AWS, Azure, GCP.
Private Cloud	Dedicated infrastructure for one organisation. Higher control and security.
Hybrid Cloud	Combination of public and private clouds.
Community Cloud	Shared among organisations with common compliance needs.

AWS global infrastructure and Amazon S3. AWS offers over 200 fully featured services across compute, storage, databases, networking, analytics, machine learning and security. Its global infrastructure is summarised below.

Component	Count	Description
Regions	33+	Geographic locations where AWS clusters data centres.
Availability Zones	105+	Isolated data centre clusters within each region for fault tolerance.
Edge Locations	400+	CloudFront CDN endpoints for low-latency content delivery.
Countries Served	190+	AWS services available globally.

Amazon Simple Storage Service (S3) is an object storage service that provides industry-leading scalability, data availability, security and performance. S3 stores data as objects within flat containers called buckets and is designed for 99.999999999% (eleven nines) durability by redundantly storing data across multiple devices in multiple facilities.

Concept	Description
Bucket	A globally unique named container for storing S3 objects.
Object	Any file stored in S3 (HTML, CSS, images). Up to 5 TB per object.
Key	The unique identifier (path/name) for an object within a bucket.
Region	AWS geographic location where the bucket resides.
Bucket Policy	JSON document defining access permissions for the bucket.
ACL	Access Control List – object-level permissions for individual files.
Versioning	Maintains multiple historical versions of an object.
Endpoint	A URL generated by AWS for accessing the static website hosted in a bucket.

AWS free tier limits relevant to this project

Service	Limit	Description
S3 Storage	5 GB	Standard storage per month
GET Requests	20,000	HTTP GET / LIST requests per month
PUT Requests	2,000	HTTP PUT / COPY / POST / DELETE requests
Data Transfer OUT	15 GB	Data transferred out of S3 to the internet per month

The CloudStore website is well within these limits. A typical static website of 50 KB HTML/CSS stays free for millions of page views per month.

1.4 Literature Review

The following academic and industry literature establishes the theoretical foundation and justifies the technology choices made in this project.

- **Mell & Grance (2011) – NIST Definition:** the foundational definition of cloud computing, identifying five essential characteristics, three service models and four deployment models.
- **Armbrust et al. (2010) – A View of Cloud Computing:** identified ten key obstacles and opportunities for cloud adoption, highlighting static content hosting as a primary low-cost, high-availability use case.
- **Buyya et al. (2009) – Market-Oriented Cloud Architecture:** demonstrated the economic advantages of cloud-based resource provisioning over traditional hosting, aligning with S3's pay-per-use pricing.

- **Amazon Web Services (2024) – S3 Documentation:** technical specifications for bucket configuration, access control, versioning, lifecycle policies and static website hosting.
- **W3Techs (2024):** Amazon S3 is used by over 28% of all websites for content storage or delivery.
- **Gartner (2023):** cloud adoption continues to grow at approximately 20% annually, with static hosting and serverless architectures among the fastest-growing segments for small and medium enterprises.
- **Hashizume et al. (2013) – Cloud Security Analysis:** analysed cloud security vulnerabilities and best practices; the public bucket policy in this project was informed by its access control principles.

The literature collectively confirms that AWS S3 is a technically sound, economically viable and widely adopted solution for static website hosting – making it an appropriate choice for this internship project.

2. COMPANY PROFILE

About the Company:

DLK Software Solutions is a software company based in Chennai that has been operating since 2012. The company is **ISO 9001** certified and **MSME registered**, and it offers software development training and internship programs that give students practical exposure to industry technologies.

As part of my internship at DLK Software Solutions, I worked in the domain of **Cloud Computing** from 6th June 2026 to 6th July 2026 under the guidance of **Mr. Velmurugan, Project Team Head**. During the internship I designed the CloudStore static website using HTML5 and CSS3 and deployed it on **AWS S3**, covering bucket creation, file upload, bucket policy configuration, static website hosting and testing.

The internship provided valuable practical knowledge of AWS S3 configuration, JSON bucket policies, IAM access control, static hosting and cloud deployment workflows. It helped me apply the theoretical concepts of cloud computing in a real working environment and strengthened the foundation for exploring advanced AWS services such as EC2, Lambda, CloudFront, RDS and DynamoDB.

Company Details:

- **Company Name:** DLK Software Solutions
- **Established:** Since 2012

- **Registrations:** ISO 9001 | MSME Registered
- **Location:** Vadapalani, Chennai – 600 026, Tamil Nadu, India
- **Internship Domain:** Cloud Computing
- **Internship Duration:** 6th June 2026 – 6th July 2026
- **Internship Guide:** Mr. Velmurugan, Project Team Head
- **Project Completed During Internship:** CloudStore – Static Website Deployment using AWS S3 Cloud Storage

Technologies Used During the Internship:

- **Languages:** HTML5, CSS3
- **Cloud Platform:** Amazon Web Services (AWS) – Amazon S3 and the AWS Management Console
- **Region:** Asia Pacific (Mumbai) – ap-south-1
- **Development Tool:** Visual Studio Code
- **Testing Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge
- **Access Control:** JSON bucket policy
- **Evaluation Mirror:** GitHub Pages

3. DAILY WORK REPORT

Day 1: Reported at DLK Software Solutions, Chennai. Attended the orientation session and was briefed on the internship guidelines, the Cloud Computing domain and the reporting format. Met the mentor, Mr. Velmurugan, Project Team Head.

Day 2: Studied the fundamentals of cloud computing – the NIST definition, the five essential characteristics and the service models (IaaS, PaaS and SaaS).

Day 3: Studied the cloud deployment models (public, private, hybrid and community cloud) and the benefits of cloud computing.

Day 4: Studied the AWS global infrastructure – regions, availability zones and edge locations – and the AWS service portfolio.

Day 5: Created a free-tier AWS account (email verification and phone OTP) and explored the AWS Management Console.

Day 6: Studied Amazon S3 concepts – buckets, objects, keys, regions, bucket policies, ACLs, versioning and website endpoints.

Day 7: Studied the key S3 features (durability, availability, lifecycle rules, event triggers, encryption, access logging) and the AWS free tier limits.

Day 8: Carried out a literature review (NIST, Armbrust et al., Buyya et al., AWS documentation) and finalised the project topic with the mentor.

Day 9: Prepared the problem statement, scope and objectives of the project. Listed the hardware and software requirements.

Day 10: Designed the system architecture and data flow (Browser → DNS → S3 endpoint → bucket policy → index.html) and prepared the block diagram.

Day 11: Planned the structure of the CloudStore website – seven sections, colour palette and fonts.

Day 12: Set up Visual Studio Code and created index.html. Developed the navigation bar and the hero section.

Day 13: Developed the statistics bar and the features section using CSS Grid.

Day 14: Developed the "How It Works" and pricing sections of the website.

Day 15: Developed the call-to-action section and footer. Added hover animations and responsive media queries.

Day 16: Tested every section of the website locally in Chrome, Firefox and Edge and fixed layout issues.

Day 17: Simulated mobile screen sizes (375 px width) using Chrome DevTools and verified the responsive layout.

Day 18: Created the S3 bucket **cloudstore-swathi-2026** in the Asia Pacific (Mumbai) region with SSE-S3 default encryption.

Day 19: Disabled Block Public Access and confirmed the object ownership setting (ACLs disabled – bucket owner enforced).

Day 20: Uploaded index.html to the bucket and verified the file in the object list.

Day 21: Studied the structure of a bucket policy and wrote the public-read policy (PublicReadGetObject).

Day 22: Applied the bucket policy and verified the public access status of the bucket.

Day 23: Enabled static website hosting, set index.html as the index document and obtained the website endpoint URL.

Day 24: Opened the S3 endpoint URL in Chrome, Firefox and Edge and verified that the website loads correctly.

Day 25: Studied the S3 security features (IAM, encryption, logging, MFA Delete) and CloudFront with Origin Access Control for production use.

Day 26: Documented the advantages and limitations of S3 static hosting and confirmed that usage stayed within the AWS free tier.

Day 27: Published an identical mirror of the website on GitHub Pages for easy access during evaluation.

Day 28: Executed all 15 test cases – all passed.

Day 29: Captured screenshots and prepared the analysis, conclusion, future work and reference sections.

Day 30: Compiled and reviewed the final internship report and submitted it to the mentor.

4. BLOCK DIAGRAM

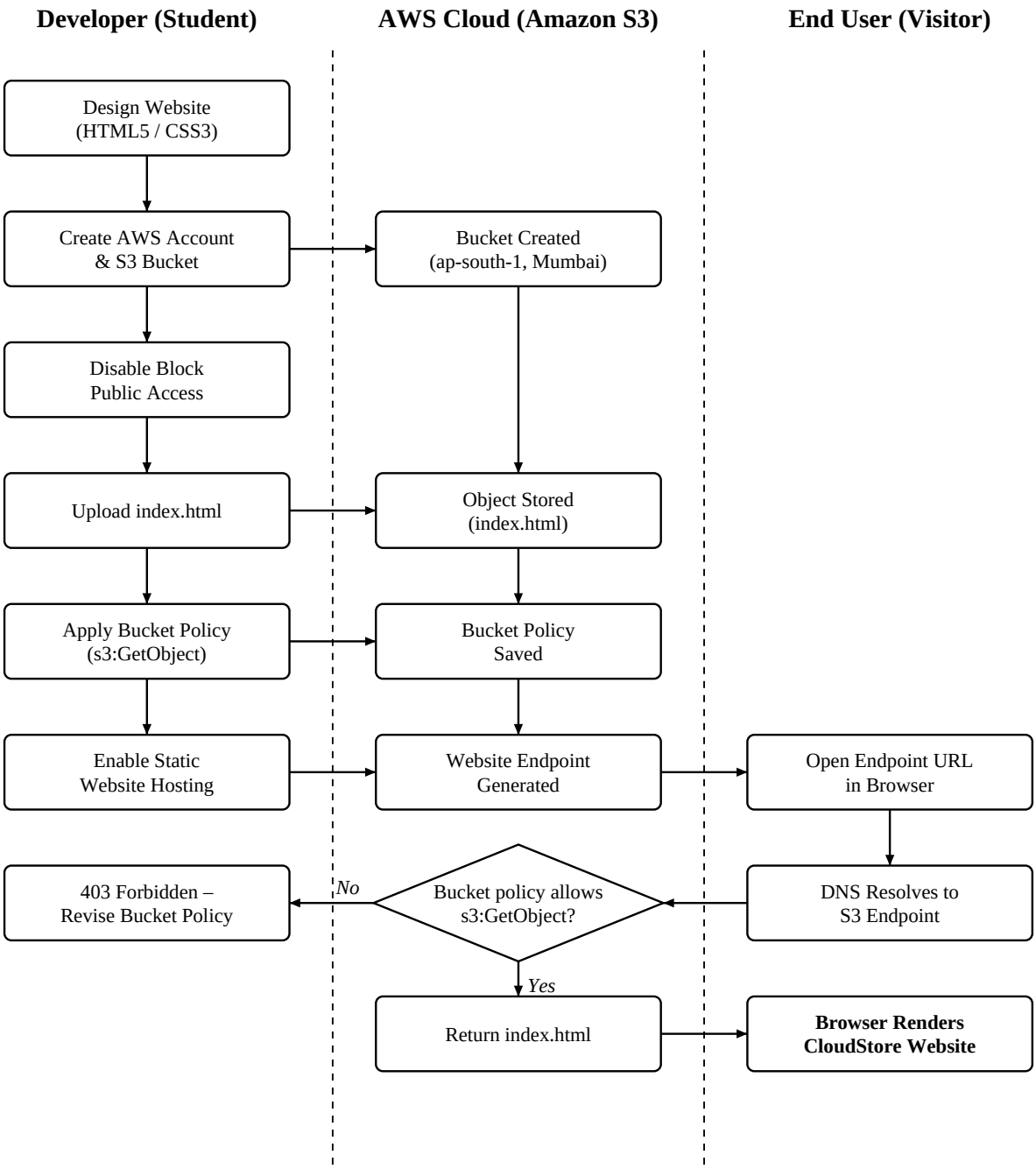


Figure 4.1: Block diagram of the CloudStore deployment and website access workflow

5. CLOUDSTORE WEBSITE HOSTING SYSTEM

5.1 Existing System

Most traditional website hosting setups depend on a dedicated or virtual server that has to be purchased or leased, configured and maintained by the owner. The owner must configure the server operating system, install web server software, manage SSL certificates, and handle hardware failures and software updates. This approach suffers from several limitations:

- **Server setup and configuration** – The operating system and web server software have to be installed and configured before even a simple page can be published.
- **Maintenance overhead** – Hardware failures and software updates must be handled continuously by the owner.
- **SSL certificate management** – Certificates have to be obtained, installed and renewed manually.
- **High cost** – Purchasing or leasing servers requires significant financial investment, which is unnecessary for small projects and demonstrations.
- **Complexity for beginners** – The process requires technical expertise and time, making it difficult for students and first-time developers.
- **Limited scalability** – A fixed server cannot automatically handle a sudden increase in traffic.

These drawbacks make traditional hosting too expensive, too complex or too limited in scalability for demonstration and academic purposes. Therefore, there is a need for a simple, low-cost and highly available hosting method for static content.

5.2 Proposed System

The proposed system hosts the **CloudStore** static website directly from an **Amazon S3** bucket. The website is built with **HTML5 and CSS3**, stored as an object in the bucket and served through the S3 static website endpoint. Key aspects of the proposed system include:

- **Zero server management** – There is no need to install, configure or patch web server software.
- **High durability and availability** – S3 offers 99.999999999% (11 nines) durability and a 99.99% availability SLA for the standard storage class.
- **Automatic scalability** – S3 automatically handles a large number of concurrent requests.
- **Low cost** – The project stays within the AWS free tier, so no billing charges are incurred.
- **Controlled public access** – A JSON bucket policy grants read-only (s3:GetObject) access to visitors.

- **Rapid deployment** – A website can be live in under 20 minutes using only the AWS web console.
- **Evaluation mirror** – An identical copy of the same build is also published on GitHub Pages so that the coordinator can access the site without an AWS session.

By using S3 static hosting, the proposed system removes server management, reduces cost and provides a professional, globally accessible deployment suitable for internship demonstrations and small projects.

5.3 Capabilities

The CloudStore website and the S3 hosting setup offer the following key capabilities. The website is organised into the sections below:

Section	Description
Navigation Bar	Sticky, translucent top bar with CloudStore branding, links to Features, How it Works and Pricing, and a 'Get Started Free' button.
Hero Section	Gradient-tinted section with the tag 'Powered by AWS S3', the headline 'Store, Access & Share Files From Anywhere', a subtitle and two call-to-action buttons.
Stats Bar	Dark navy strip showing four metrics: 99.9% Uptime, 256-bit AES Encryption, 150+ Countries and 10M+ Files Stored.
Features Section	Six cards: Bank-Grade Security, Lightning Fast Upload, Global CDN Access, Easy File Sharing, Folder Organisation and Auto Backup.
How It Works	Four numbered steps: Create Account, Upload Files, Organise and Access Anywhere.
Pricing	Three plan cards – Free (₹0), Pro (₹299/month, Most Popular) and Business (₹999/month) – each listing plan features.
CTA Section	Dark gradient closing section with a final call-to-action button.
Footer	Dark footer crediting the author and DLK Software Solutions Internship Project, 2026.

Key AWS S3 capabilities used or available in the project:

- **Static Hosting** – Built-in feature to serve HTML/CSS/JS as a website via a public endpoint.
- **High Durability and Availability** – Data replicated across multiple Availability Zones.
- **Versioning** – Keeps multiple versions of objects for rollback and audit.
- **Lifecycle Rules** – Automate transitions to cheaper storage tiers or deletion.
- **Event Triggers** – Invoke AWS Lambda or SNS on object upload, modification or deletion.
- **Encryption** – Server-side and client-side encryption options.
- **Access Logging** – Logs all bucket access requests for auditing.

6. PROPOSED METHOD

6.1 System Requirements

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher / AMD Ryzen 3 or higher
RAM	Minimum 2 GB (4 GB recommended)
Storage	Minimum 100 MB free space for project files
Internet	Stable broadband connection (minimum 2 Mbps)
Display	Minimum 1024 x 768 screen resolution

Software Requirements

Software	Details
Operating System	Windows 10/11, macOS 11+, or Linux Ubuntu 20.04+
Web Browser	Google Chrome 110+ / Mozilla Firefox 110+ / Microsoft Edge 110+
Text Editor	Visual Studio Code (VS Code) – recommended
AWS Account	Free tier AWS account – created at aws.amazon.com
AWS Console	Web-based management console
Languages	HTML5 and CSS3 – no compiler or runtime installation required

6.2 System Architecture

The project follows a three-tier cloud delivery model: the **client tier** (user browser), the **transport tier** (internet/DNS) and the **hosting tier** (AWS S3 bucket). This lightweight, server-less architecture is ideal for static content delivery.

Component	Role
User Browser	Sends HTTP requests to the S3 website endpoint URL.
DNS Resolution	AWS Route 53 or public DNS resolves the S3 endpoint to the correct region.
AWS S3 Bucket	Stores and serves HTML/CSS files; hosts the static website via the built-in endpoint.
Bucket Policy	JSON policy granting s3:GetObject permission to all public users.
S3 Endpoint	Auto-generated public URL: [bucket].s3-website.[region].amazonaws.com
CloudFront (optional)	CDN for HTTPS support and edge caching across 400+ global locations.

Data flow: User types URL → DNS resolves to the S3 endpoint → S3 checks the bucket policy → returns index.html → browser renders the CloudStore website.

6.3 Website Design and Development

The CloudStore website was designed and built using HTML5 and CSS3 in a single index.html file, using a modern light-theme aesthetic with a navy/blue accent palette commonly used in technology SaaS products. The page contains seven sections: navigation, hero, statistics bar, features, how-it-works, pricing and footer.

- Single-file deployment (index.html) for simplicity of S3 upload and management.
- Light background (#F0F4FF) with navy (#0B1D3A) headings for a clean SaaS look.
- CSS Grid for responsive feature cards, steps and pricing plan layouts.
- Blue-to-cyan gradient accents (#1A56DB → #06B6D4) for call-to-action buttons and highlights.
- Hover animations on cards for an interactive user experience.
- Google Fonts (Syne and DM Sans) for headings and body text.

6.4 S3 Bucket Creation

An AWS free tier account was created at aws.amazon.com. After email verification and phone OTP, the AWS Management Console was accessible. The S3 service was located using the search bar in the top navigation, and the bucket was created with the following settings:

Setting	Value
Bucket Name	cloudstore-swathi-2026 (globally unique)
AWS Region	Asia Pacific (Mumbai) – ap-south-1
Block Public Access	DISABLED – to allow public website access
Bucket Versioning	Disabled (not required for static hosting)
Default Encryption	Amazon S3-managed keys (SSE-S3)
Object Ownership	ACLs disabled – bucket owner enforced

6.5 File Upload

- Open the cloudstore-swathi-2026 bucket in the AWS S3 console.
- Click ‘Upload’ to open the upload interface.
- Click ‘Add files’ and select the index.html file.
- Leave the storage class as ‘Standard’ (default).
- Click ‘Upload’ and wait for the progress bar to reach 100%.
- The file appears in the bucket object list with its size, date and storage class.

6.6 Bucket Policy Configuration

A JSON bucket policy was applied to grant anonymous public read access to all objects in the bucket, allowing the website to be accessed by any internet user without authentication.

Field	Value	Explanation
Version	2012-10-17	Policy language version
Sid	PublicReadGetObject	Statement ID – descriptive label
Effect	Allow	Whether to allow or deny the action
Principal	*	Who the policy applies to – everyone
Action	s3:GetObject	The S3 operation permitted – read/download
Resource	arn:aws:s3:::bucket-name/*	Which objects the policy applies to

6.7 Static Website Hosting

- Navigate to the bucket → Properties tab.
- Scroll to the ‘Static website hosting’ section.
- Click ‘Edit’ and select ‘Enable’.
- Enter ‘index.html’ in the Index document field.
- Optionally enter ‘error.html’ in the Error document field.
- Click ‘Save changes’.

The website endpoint URL is then displayed:

http://cloudstore-swathi-2026.s3-website.ap-south-1.amazonaws.com

To make the project easy to access for the coordinator/evaluator without relying on an AWS session, an identical copy of the same design was also published to GitHub Pages at https://swathikumaravel48-tech.github.io/cloudstore. The AWS S3 deployment described above remains the primary demonstration of the cloud computing concepts covered in this internship.

6.8 Security in AWS S3

Security is a critical aspect of any cloud deployment. While this project uses a public bucket policy for website hosting, understanding the broader security features of AWS S3 is essential for professional cloud development.

Security Feature	Description
Bucket Policies	JSON-based policies controlling access at bucket level.
IAM Policies	Control which AWS users/roles can interact with S3.
Block Public Access	AWS default that blocks all public access; disabled here for hosting.
Server-Side Encryption	AWS S3 encrypts all objects at rest using AES-256 by default.
SSL/TLS in Transit	Data transfer encrypted via HTTPS (requires CloudFront for website endpoints).
Access Logging	Logs all access requests to a separate bucket for audit.
Versioning	Maintains object history for rollback protection.
MFA Delete	Requires multi-factor authentication to permanently delete objects.

For production websites, it is strongly recommended to use CloudFront with an Origin Access Control (OAC) instead of a public bucket policy, keeping the S3 bucket private while serving content securely over HTTPS.

7. ALGORITHM

7.1 Website Development – Algorithm

1. Plan the sections of the website (navigation, hero, statistics, features, how-it-works, pricing, footer).
2. Create a single index.html file in Visual Studio Code.
3. Define the colour palette, fonts and layout using CSS variables and CSS Grid.
4. Develop each section of the page using HTML5 and CSS3.
5. Add hover animations and responsive media queries.
6. Open the page locally in Chrome, Firefox and Edge and verify every section.
7. Simulate a 375 px mobile screen and confirm the layout adapts correctly.

7.2 S3 Bucket Creation – Algorithm

1. Sign in to the AWS Management Console and open the S3 service.
2. Click ‘Create bucket’ and enter a globally unique bucket name.
3. Select the AWS Region – Asia Pacific (Mumbai) ap-south-1.
4. Disable ‘Block all public access’ and acknowledge the warning.
5. Keep versioning disabled and choose SSE-S3 as the default encryption.
6. Confirm that ACLs are disabled (bucket owner enforced).
7. Click ‘Create bucket’ and verify the bucket in the bucket list.

7.3 File Upload – Algorithm

1. Open the bucket in the S3 console.
2. Click ‘Upload’ and then ‘Add files’.
3. Select index.html from the local system.
4. Keep the storage class as Standard.
5. Click ‘Upload’ and wait until the progress reaches 100%.
6. Verify that index.html appears in the object list.

7.4 Bucket Policy Configuration – Algorithm

1. Open the bucket → Permissions tab → Bucket policy → Edit.
2. Enter the JSON policy with Effect Allow, Principal *, Action s3:GetObject.
3. Set the Resource to arn:aws:s3:::cloudstore-swathi-2026/*.
4. Click ‘Save changes’.
5. Confirm that no error is shown and the bucket is marked as publicly accessible.

7.5 Static Website Hosting – Algorithm

1. Open the bucket → Properties tab.
2. Scroll to ‘Static website hosting’ and click ‘Edit’.
3. Select ‘Enable’.
4. Enter index.html as the Index document (and optionally error.html as the Error document).
5. Click ‘Save changes’.
6. Copy the generated website endpoint URL.

7.6 Website Access (Request Flow) – Algorithm

1. The visitor enters the website endpoint URL in the browser.
2. DNS resolves the S3 website endpoint to the correct AWS region.
3. The request reaches the S3 bucket.
4. S3 checks the bucket policy for the s3:GetObject permission.
5. If the permission is not granted, S3 returns an access denied (403) response.
6. If the permission is granted, S3 returns the index.html object.
7. The browser renders the CloudStore website.

8. CODE IMPLEMENTATION

8.1 Bucket Policy

Grants anonymous public read access to every object in the bucket so that the website can be viewed by any visitor.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::cloudstore-swathi-2026/*"
    }
  ]
}
```

8.2 Website Source Code (index.html)

The complete HTML5/CSS3 source code of the CloudStore static website, as deployed on AWS S3 (and mirrored on GitHub Pages), is given below. Emoji characters in the listing are shown as HTML character references.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>CloudStore — Secure Cloud Storage</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link
    href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@400;500;600&display=swap"
     rel="stylesheet">
<style>
  :root{
    --bg: #F0F4FF;
    --navy: #0B1D3A;
    --blue: #1A56DB;
    --sky: #3B82F6;
    --accent: #06B6D4;
    --white: #FFFFFF;
    --text: #1E293B;
    --muted: #64748B;
    --card: #FFFFFF;
    --border: #DBEAFE;
    --radius: 16px;
  }
  *{ box-sizing:border-box; margin:0; padding:0; }
  html{ scroll-behavior:smooth; }
  body{
    background: var(--bg);
    color: var(--text);
    font-family: 'DM Sans', sans-serif;
    line-height: 1.65;
  }
  h1,h2,h3,h4{ font-family:'Syne',sans-serif; }
  a{ text-decoration:none; color:inherit; }
  ::selection{ background:var(--accent); color:var(--white); }

  .wrap{ max-width:1080px; margin:0 auto; padding:0 28px; }

  /* ---- NAV ---- */
  nav{
    background:rgba(255,255,255,0.85);
    backdrop-filter:blur(12px);
    border-bottom:1px solid var(--border);
```

```
  position:sticky; top:0; z-index:50;
}
nav .inner{
  display:flex; align-items:center;
  justify-content:space-between;
  padding:16px 28px; max-width:1080px; margin:0 auto;
}
.logo{ font-family:'Syne',sans-serif; font-weight:800; font-size:20px; color:var(--navy); }
.logo span{ color:var(--blue); }
.nav-links{ display:flex; gap:28px; font-size:14.5px; color:var(--muted); }
.nav-links a:hover{ color:var(--blue); }
.nav-btn{
  background:var(--blue); color:#fff;
  padding:9px 20px; border-radius:99px;
  font-size:14px; font-weight:600;
}
.nav-btn:hover{ background:#1244b8; }
@media(max-width:640px){ .nav-links{display:none;} }

/* ---- HERO ---- */
.hero{
  padding:90px 0 70px;
  text-align:center;
  background: linear-gradient(180deg, #EEF2FF 0%, #F0F4FF 100%);
}
.hero-tag{
  display:inline-block;
  background:#DBEAFE; color:var(--blue);
  font-size:13px; font-weight:600;
  padding:6px 16px; border-radius:99px;
  margin-bottom:22px; letter-spacing:.03em;
}
.hero h1{
  font-size:clamp(34px,6vw,64px);
  font-weight:800;
  color:var(--navy);
  line-height:1.1;
  max-width:780px;
  margin:0 auto 22px;
}
.hero h1 span{
  background: linear-gradient(90deg, var(--blue), var(--accent));
  -webkit-background-clip:text; -webkit-text-fill-color:transparent;
}
.hero p{
  color:var(--muted); font-size:18px;
  max-width:540px; margin:0 auto 36px;
}
.hero-btns{ display:flex; gap:14px; justify-content:center; flex-wrap:wrap; }
.btn-primary{
  background:var(--blue); color:#fff;
  padding:14px 28px; border-radius:99px;
  font-size:15px; font-weight:600;
  box-shadow:0 4px 14px rgba(26,86,219,.35);
}
.btn-primary:hover{ background:#1244b8; }
.btn-outline{
  border:2px solid var(--border);
  background:#fff; color:var(--navy);
  padding:13px 26px; border-radius:99px;
  font-size:15px; font-weight:600;
}
.btn-outline:hover{ border-color:var(--blue); color:var(--blue); }

/* ---- STATS BAR ---- */
.stats{
  background:var(--navy);
  padding:36px 0;
}
.stats-grid{
  display:grid; grid-template-columns:repeat(4,1fr);
  gap:20px; text-align:center;
}
.stat-num{
  font-family:'Syne',sans-serif;
  font-size:32px; font-weight:800;
  color:#fff; display:block;
```

```
}
.stat-lbl{ font-size:13px; color:#94A3B8; }
@media(max-width:640px){
  .stats-grid{ grid-template-columns:repeat(2,1fr); }
}

/* ---- FEATURES ---- */
.features{ padding:88px 0; }
.section-label{
  font-size:13px; font-weight:600; letter-spacing:.06em;
  color:var(--blue); text-transform:uppercase;
  display:block; margin-bottom:12px;
}
.section-title{
  font-size:clamp(24px,4vw,38px);
  font-weight:800; color:var(--navy);
  max-width:520px; margin-bottom:52px;
}
.feat-grid{
  display:grid;
  grid-template-columns:repeat(3,1fr);
  gap:24px;
}
@media(max-width:760px){ .feat-grid{ grid-template-columns:1fr; } }
.feat-card{
  background:var(--card);
  border:1px solid var(--border);
  border-radius:var(--radius);
  padding:30px;
  transition:transform .2s, box-shadow .2s;
}
.feat-card:hover{
  transform:translateY(-4px);
  box-shadow:0 12px 32px rgba(26,86,219,.1);
}
.feat-icon{
  width:48px; height:48px; border-radius:12px;
  background:linear-gradient(135deg,#DBEAFE,#EFF6FF);
  display:flex; align-items:center; justify-content:center;
  font-size:22px; margin-bottom:18px;
}
.feat-card h3{ font-size:18px; margin-bottom:8px; color:var(--navy); }
.feat-card p{ font-size:14.5px; color:var(--muted); }

/* ---- HOW IT WORKS ---- */
.how{ background:#fff; padding:88px 0; }
.steps{ display:grid; grid-template-columns:repeat(4,1fr); gap:28px; }
@media(max-width:760px){ .steps{ grid-template-columns:repeat(2,1fr); } }
@media(max-width:480px){ .steps{ grid-template-columns:1fr; } }
.step{ text-align:center; }
.step-num{
  width:52px; height:52px; border-radius:50%;
  background:linear-gradient(135deg,var(--blue),var(--accent));
  color:#fff; font-family:'Syne',sans-serif;
  font-size:20px; font-weight:800;
  display:flex; align-items:center; justify-content:center;
  margin:0 auto 16px;
}
.step h4{ font-size:16px; margin-bottom:8px; color:var(--navy); }
.step p{ font-size:14px; color:var(--muted); }

/* ---- PRICING ---- */
.pricing{ padding:88px 0; }
.plans{ display:grid; grid-template-columns:repeat(3,1fr); gap:24px; }
@media(max-width:760px){ .plans{ grid-template-columns:1fr; } }
.plan{
  background:#fff; border:1.5px solid var(--border);
  border-radius:var(--radius); padding:32px 28px;
}
.plan.popular{
  border-color:var(--blue);
  box-shadow:0 8px 32px rgba(26,86,219,.18);
  position:relative;
}
.popular-tag{
  position:absolute; top:-14px; left:50%; transform:translateX(-50%);
  background:var(--blue); color:#fff;
```

```
    font-size:12px; font-weight:700;
    padding:5px 16px; border-radius:99px;
    white-space:nowrap;
  }
  .plan-name{ font-family:'Syne',sans-serif; font-size:18px; font-weight:700; color:var(--navy);
     margin-bottom:8px; }
  .plan-price{
    font-family:'Syne',sans-serif;
    font-size:38px; font-weight:800;
    color:var(--navy); margin:14px 0 6px;
  }
  .plan-price span{ font-size:16px; font-weight:500; color:var(--muted); }
  .plan p{ font-size:14px; color:var(--muted); margin-bottom:24px; }
  .plan ul{ list-style:none; margin-bottom:28px; }
  .plan ul li{
    font-size:14.5px; color:var(--text);
    padding:7px 0; border-bottom:1px solid var(--border);
    display:flex; align-items:center; gap:10px;
  }
  .plan ul li::before{ content:"✓"; color:var(--blue); font-weight:700; }
  .plan-btn{
    display:block; text-align:center;
    padding:12px; border-radius:99px;
    font-weight:600; font-size:14.5px;
    background:var(--bg); color:var(--navy);
    border:1.5px solid var(--border);
  }
  .plan.popular .plan-btn{
    background:var(--blue); color:#fff; border:none;
  }
  .plan-btn:hover{ filter:brightness(.95); }

  /* ---- CTA ---- */
  .cta-section{
    background:linear-gradient(135deg,var(--navy) 0%,#0F3460 100%);
    padding:80px 0; text-align:center;
  }
  .cta-section h2{
    font-size:clamp(26px,4vw,42px);
    font-weight:800; color:#fff; margin-bottom:14px;
  }
  .cta-section p{ color:#94A3B8; font-size:17px; margin-bottom:32px; }

  /* ---- FOOTER ---- */
  footer{
    background:var(--navy);
    border-top:1px solid #1E3A5F;
    padding:32px 0; text-align:center;
  }
  footer p{ font-size:13.5px; color:#64748B; }
  footer span{ color:var(--accent); }
</style>
</head>
<body>

<!-- NAV -->
<nav>
  <div class="inner">
    <div class="logo">Cloud<span>Store</span></div>
    <div class="nav-links">
      <a href="#features">Features</a>
      <a href="#how">How it Works</a>
      <a href="#pricing">Pricing</a>
    </div>
    <a class="nav-btn" href="#pricing">Get Started Free</a>
  </div>
</nav>

<!-- HERO -->
<section class="hero">
  <div class="wrap">
    <div class="hero-tag">&#x2601;&#xFE0F; Powered by AWS S3</div>
    <h1>Store, Access &amp; Share Files <span>From Anywhere</span></h1>
    <p>CloudStore gives you secure, fast, and reliable cloud storage — accessible from any device, any
       time.</p>
    <div class="hero-btns">
      <a class="btn-primary" href="#pricing">Start Free — No Card Needed</a>
```

```
      <a class="btn-outline" href="#how">See How It Works</a>
    </div>
  </div>
</section>

<!-- STATS -->
<div class="stats">
  <div class="wrap">
    <div class="stats-grid">
      <div>
        <span class="stat-num">99.9%</span>
        <span class="stat-lbl">Uptime Guaranteed</span>
      </div>
      <div>
        <span class="stat-num">256-bit</span>
        <span class="stat-lbl">AES Encryption</span>
      </div>
      <div>
        <span class="stat-num">150+</span>
        <span class="stat-lbl">Countries Supported</span>
      </div>
      <div>
        <span class="stat-num">10M+</span>
        <span class="stat-lbl">Files Stored</span>
      </div>
    </div>
  </div>
</div>

<!-- FEATURES -->
<section class="features" id="features">
  <div class="wrap">
    <span class="section-label">Why CloudStore</span>
    <h2 class="section-title">Everything you need to store files with confidence</h2>
    <div class="feat-grid">
      <div class="feat-card">
        <div class="feat-icon">&#x1F512;</div>
        <h3>Bank-Grade Security</h3>
        <p>Your files are encrypted end-to-end with 256-bit AES encryption — same standard as banking
            apps.</p>
      </div>
      <div class="feat-card">
        <div class="feat-icon">&#x26A1;</div>
        <h3>Lightning Fast Upload</h3>
        <p>Multi-part upload technology means even large files transfer in seconds across all
            regions.</p>
      </div>
      <div class="feat-card">
        <div class="feat-icon">&#x1F30D;</div>
        <h3>Global CDN Access</h3>
        <p>Files are cached at edge locations worldwide — so access is fast no matter where you are.</p>
      </div>
      <div class="feat-card">
        <div class="feat-icon">&#x1F517;</div>
        <h3>Easy File Sharing</h3>
        <p>Generate secure shareable links with optional expiry — share files without sharing your
            password.</p>
      </div>
      <div class="feat-card">
        <div class="feat-icon">&#x1F4C1;</div>
        <h3>Folder Organisation</h3>
        <p>Create unlimited folders, move files around, and keep your storage clean and structured.</p>
      </div>
      <div class="feat-card">
        <div class="feat-icon">&#x1F504;</div>
        <h3>Auto Backup</h3>
        <p>Connect your device for automatic daily backups — never lose an important file again.</p>
      </div>
    </div>
  </div>
</section>

<!-- HOW IT WORKS -->
<section class="how" id="how">
  <div class="wrap">
    <span class="section-label">How It Works</span>
    <h2 class="section-title">Up and running in under a minute</h2>
```

```
    <div class="steps">
      <div class="step">
        <div class="step-num">1</div>
        <h4>Create Account</h4>
        <p>Sign up free with just your email — no credit card required to get started.</p>
      </div>
      <div class="step">
        <div class="step-num">2</div>
        <h4>Upload Files</h4>
        <p>Drag and drop files directly into your CloudStore dashboard from any browser.</p>
      </div>
      <div class="step">
        <div class="step-num">3</div>
        <h4>Organise</h4>
        <p>Create folders, rename files, and keep everything exactly where you want it.</p>
      </div>
      <div class="step">
        <div class="step-num">4</div>
        <h4>Access Anywhere</h4>
        <p>Log in from any device, any browser — your files are always there, always safe.</p>
      </div>
    </div>
  </div>
</section>

<!-- PRICING -->
<section class="pricing" id="pricing">
  <div class="wrap">
    <span class="section-label">Pricing</span>
    <h2 class="section-title">Simple plans, no hidden charges</h2>
    <div class="plans">
      <div class="plan">
        <div class="plan-name">Free</div>
        <div class="plan-price">₹0<span>/month</span></div>
        <p>Perfect for personal use and getting started.</p>
        <ul>
          <li>5 GB Storage</li>
          <li>File sharing links</li>
          <li>Mobile access</li>
          <li>Email support</li>
        </ul>
        <a class="plan-btn" href="#">Get Started</a>
      </div>
      <div class="plan popular">
        <div class="popular-tag">Most Popular</div>
        <div class="plan-name">Pro</div>
        <div class="plan-price">₹299<span>/month</span></div>
        <p>Great for professionals and small teams.</p>
        <ul>
          <li>100 GB Storage</li>
          <li>Password-protected links</li>
          <li>Auto backup</li>
          <li>Priority support</li>
        </ul>
        <a class="plan-btn" href="#">Start Free Trial</a>
      </div>
      <div class="plan">
        <div class="plan-name">Business</div>
        <div class="plan-price">₹999<span>/month</span></div>
        <p>For teams and businesses that need more power.</p>
        <ul>
          <li>1 TB Storage</li>
          <li>Team collaboration</li>
          <li>Advanced analytics</li>
          <li>24/7 dedicated support</li>
        </ul>
        <a class="plan-btn" href="#">Contact Sales</a>
      </div>
    </div>
  </div>
</section>

<!-- CTA -->
<section class="cta-section">
  <div class="wrap">
    <h2>Ready to store smarter?</h2>
    <p>Join thousands of users who trust CloudStore for their files.</p>
```

```
    <a class="btn-primary" href="#pricing">Create Free Account →</a>
  </div>
</section>

<!-- FOOTER -->
<footer>
  <div class="wrap">
    <p>© 2026 <span>CloudStore</span> — Built &amp;amp; deployed on AWS S3 by Swathi K | DLK Software
        Solutions Internship Project</p>
  </div>
</footer>

</body>
</html>
```

9. SAMPLE OUTPUT

9.1 Home Page

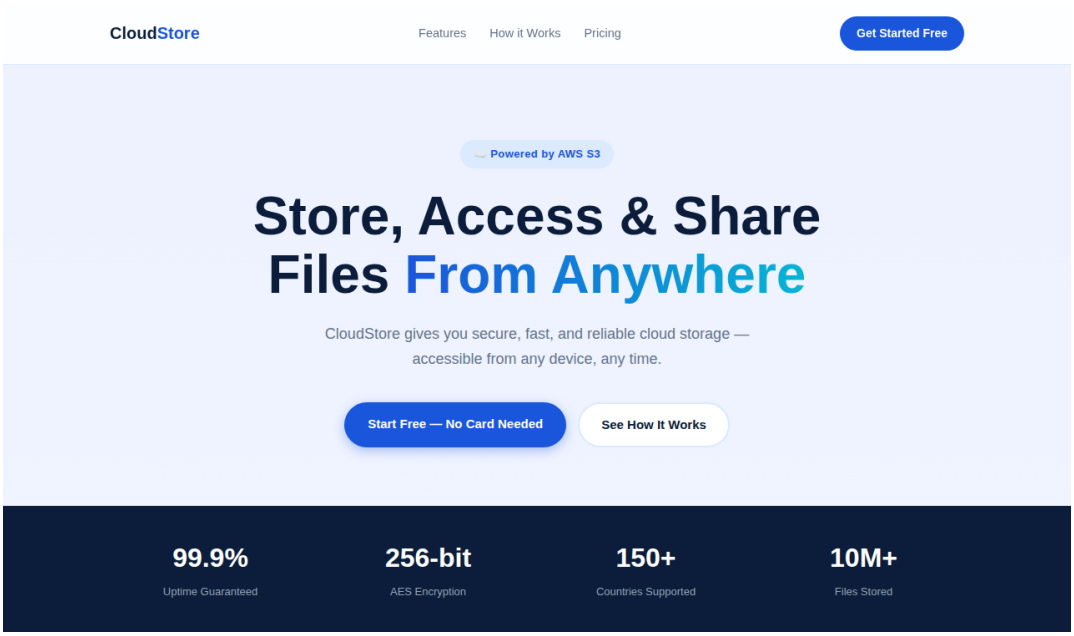


Figure 9.1: Navigation bar, hero section and statistics bar of the CloudStore website

9.2 Features Section

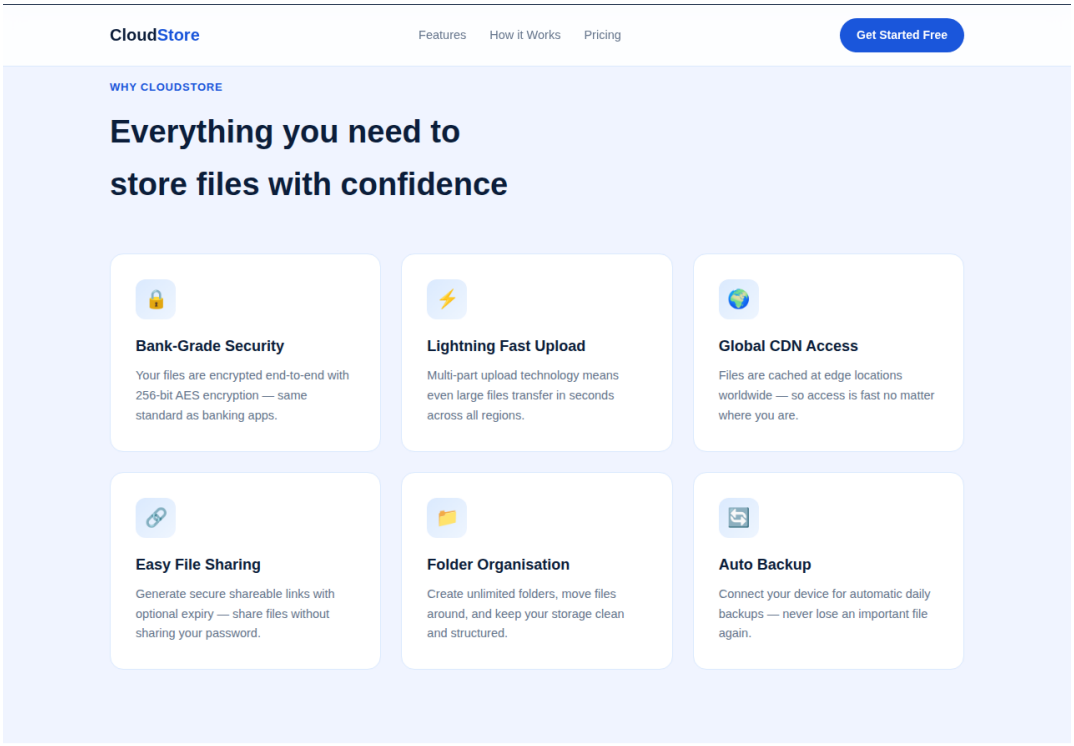


Figure 9.2: Features section with six feature cards

9.3 How It Works Section

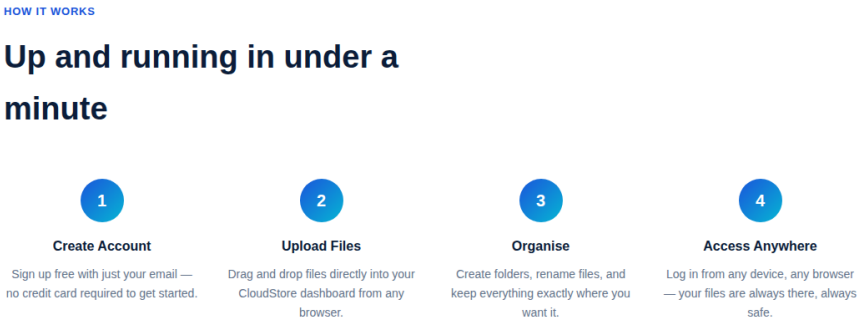


Figure 9.3: Four-step “How It Works” section

9.4 Pricing Section

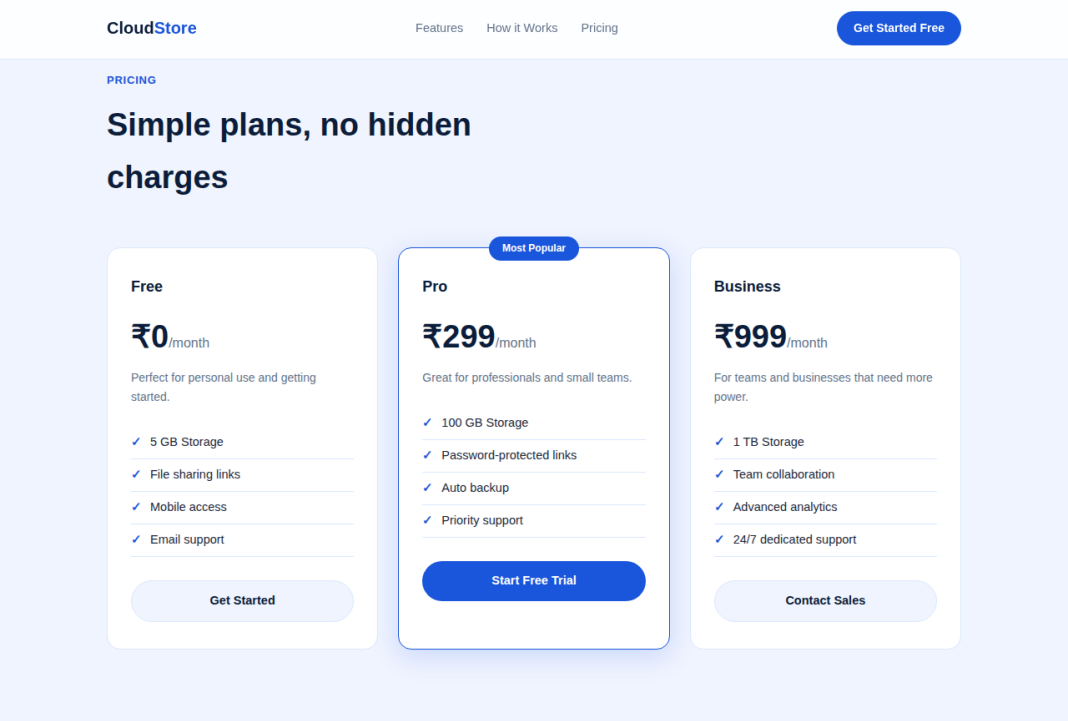


Figure 9.4: Pricing section with Free, Pro and Business plans

9.5 Call-to-Action and Footer

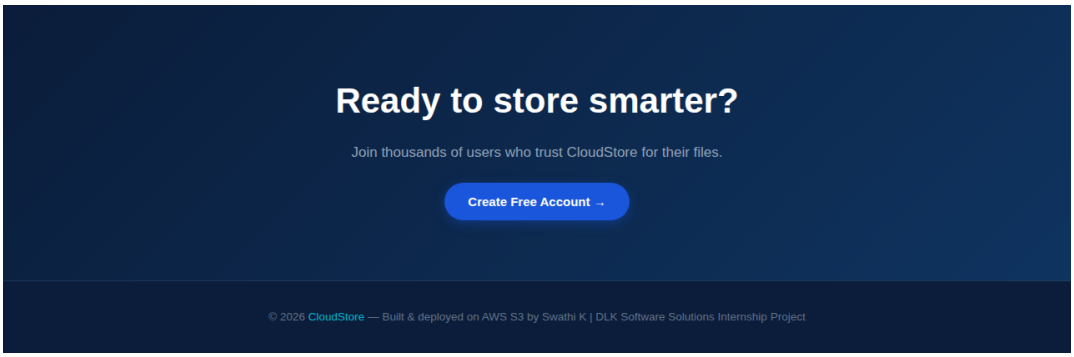


Figure 9.5: Closing call-to-action section and footer

9.6 Mobile View

The layout adapts to a 375 px wide mobile screen, with the navigation links hidden and the cards stacked in a single column.

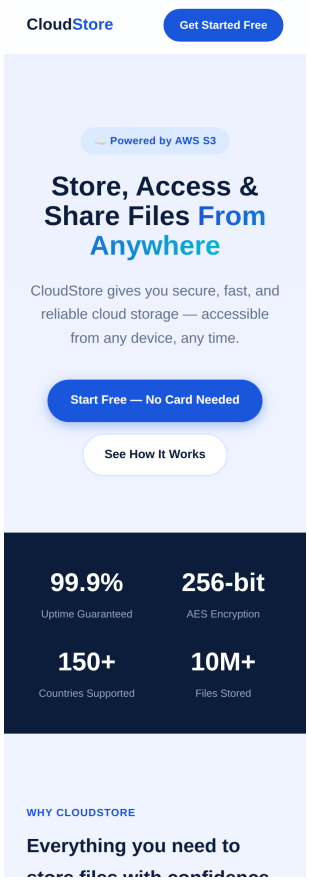


Figure 9.6: CloudStore website in the mobile view (375 px width)

9.7 Live Website Details

Field	Details
Bucket Name	cloudstore-swathi-2026
AWS Region	ap-south-1 (Asia Pacific – Mumbai)
Storage Class	S3 Standard
S3 Website Endpoint	http://cloudstore-swathi-2026.s3-website.ap-south-1.amazonaws.com
Live Demo Link (Evaluation Access)	https://swathikumaravel48-tech.github.io/cloudstore
File Uploaded	index.html (CloudStore static website)
Access Level	Public – accessible by anyone on the internet
Cost	Within AWS Free Tier – no charges incurred

The Live Demo Link is a like-for-like mirror of the same CloudStore build, published to GitHub Pages to guarantee stable, always-on access for the class coordinator during evaluation. The underlying implementation, testing and cloud concepts documented in this report were carried out on AWS S3.

10. ANALYSIS

Testing of the CloudStore website was carried out at each phase of the project to ensure that the website functions correctly, renders properly and is publicly accessible after deployment.

10.1 Testing Strategy

- **Unit Testing:** individual HTML sections tested in a local browser before upload.
- **Integration Testing:** the full page rendered locally in Chrome, Firefox and Edge.
- **Deployment Testing:** S3 upload verified through the AWS Console object listing.
- **Policy Testing:** bucket policy applied and public access status confirmed.
- **URL Testing:** S3 endpoint URL opened and the website verified in multiple browsers.
- **Responsiveness Testing:** Chrome DevTools used to simulate mobile screen sizes.
- **Cross-Browser Testing:** Chrome 120, Firefox 121 and Edge 120 were tested.

10.2 Test Cases and Results

TC ID	Test Case	Expected Result	Status
TC01	Open index.html in browser locally	All sections render	Pass
TC02	Verify navigation bar links	All links visible	Pass
TC03	Verify hero section CTA buttons	Both buttons display	Pass
TC04	Verify features cards layout	6 cards in grid	Pass
TC05	Verify statistics bar	4 metrics displayed	Pass
TC06	Verify how-it-works section	4 steps with numbers	Pass
TC07	Verify pricing section	3 plans with correct pricing	Pass
TC08	Upload index.html to S3 bucket	Upload 100% complete	Pass
TC09	Verify file in bucket object list	index.html listed	Pass
TC10	Apply bucket policy – save without error	Policy saved	Pass
TC11	Enable static website hosting	Endpoint URL generated	Pass
TC12	Open S3 endpoint URL in Chrome	Website loads correctly	Pass
TC13	Open S3 endpoint URL in Firefox	Website loads correctly	Pass
TC14	Open S3 endpoint URL in Edge	Website loads correctly	Pass
TC15	Mobile view simulation (375px width)	Layout adapts correctly	Pass

All 15 test cases passed successfully. No errors or failures were encountered during testing. The website is fully functional, publicly accessible and renders correctly across all tested browsers and screen sizes.

10.3 Results

- A fully functional, multi-section static website (CloudStore) was designed and built using HTML5 and CSS3.
- The website was deployed to an AWS S3 bucket in the Asia Pacific (Mumbai) region.
- Static website hosting was enabled with index.html as the default index document.
- A public bucket policy was applied, granting GetObject access to all internet users.
- All 15 test cases passed with zero failures across Chrome, Firefox and Edge.
- The website rendered correctly on both desktop (1920x1080) and mobile (375 px) screen sizes.
- Zero server configuration was required – the entire deployment was done through the AWS web console.
- The project remained entirely within the AWS free tier – no billing charges were incurred.

10.4 Advantages and Disadvantages

Advantages of using AWS S3 for static website hosting

Advantage	Description
Zero Server Management	No need to install, configure or patch web server software.
Extreme Cost Efficiency	AWS free tier provides 5 GB storage and 20K GET requests/month at no cost.
Eleven Nines Durability	99.999999999% durability across multiple availability zones.
High Availability	99.99% availability SLA.
Infinite Scalability	S3 automatically handles millions of concurrent requests.
Rapid Deployment	A website can be live in under 20 minutes.
Global Accessibility	Accessible from any device, anywhere, via a public URL.
Built-in Redundancy	Data replicated across multiple Availability Zones.
Versioning Support	Enables rollback if updates cause issues.
CDN Integration Ready	CloudFront can be added for HTTPS and global acceleration.
Custom Domain Support	AWS Route 53 enables mapping a custom domain.
Security by Default	AES-256 encryption at rest, HTTPS in transit, IAM policies.

Disadvantages and limitations

Limitation	Description
Static Content Only	S3 cannot execute server-side code (PHP, Python, Node.js).
No Database Support	Dynamic content requiring database queries cannot be served.
HTTP Only by Default	The S3 website endpoint does not support HTTPS without CloudFront.
Limited Error Handling	Only basic index/error document configuration is available.
No Server Logs by Default	Access logging must be enabled manually.
Bucket Name Constraints	Bucket names must be globally unique across all AWS accounts.
Learning Curve	The AWS Console can be overwhelming for absolute beginners.

11. CONCLUSION

This project successfully demonstrated the complete lifecycle of deploying a static website on the cloud using Amazon Web Services Simple Storage Service (**AWS S3**). From conceptualising and designing the CloudStore website in HTML5 and CSS3 to creating an S3 bucket, uploading files, configuring public access policies and enabling static website hosting – every step was executed methodically and documented in this report.

Through this internship at **DLK Software Solutions, Chennai**, a thorough practical understanding of cloud computing was developed. The theoretical study of cloud characteristics, service models, deployment models and the AWS global infrastructure was reinforced through hands-on implementation on the AWS platform.

The project demonstrated that AWS S3 is one of the most accessible, cost-effective and reliable cloud platforms available for static website hosting. The ability to deploy a globally accessible, highly available website in under 20 minutes – with zero server management and at no cost within the AWS free tier – highlights the transformative potential of cloud computing for developers at all experience levels.

The skills acquired during this internship – including **AWS S3 configuration, JSON bucket policies, IAM access control, static hosting and cloud deployment workflows** – provide a strong foundation for future exploration of more advanced cloud services such as AWS EC2, Lambda, CloudFront, RDS and DynamoDB.

In conclusion, this internship project not only fulfilled its stated objectives but also delivered a rewarding, industry-aligned learning experience that bridges the gap between academic knowledge and real-world cloud engineering practice. The CloudStore project stands as a tangible, deployable demonstration of the cloud computing concepts acquired during the internship.

12. FUTURE WORK

- **CloudFront CDN and HTTPS** (Immediate) – attach an AWS CloudFront distribution for HTTPS and CDN delivery.
- **Custom domain with Route 53** (Short-term) – map a custom domain using AWS Route 53 DNS.
- **CI/CD with GitHub Actions** (Short-term) – auto-deploy to S3 on every Git push.
- **Contact form via Lambda** (Medium-term) – add a contact form posting to AWS Lambda and SES.
- **Real file upload feature** (Medium-term) – enable browser-to-S3 uploads using pre-signed URLs.
- **User authentication via Cognito** (Long-term) – add login/signup using AWS Cognito.
- **Analytics via CloudWatch** (Long-term) – monitor traffic, errors and usage metrics.
- **Multi-language support** (Long-term) – serve localised versions of the website from S3.

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INTERNSHIP COMPLETION CERTIFICATE

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This is to certify that Ms. K. Swathi, Register No. 810023205037, pursuing B.Tech (Information Technology) at University College of Engineering (BIT Campus), Anna University, Tiruchirappalli has successfully completed the Internship in the domain of Cloud Computing at DLK Software Solutions, Chennai.

The Internship was undertaken from 6th June 2026 to 6th July 2026. During this period, she satisfactorily completed all assigned tasks and demonstrated a strong grasp of cloud computing concepts and deployment skills.

All data and information accessed during the internship period shall be treated as strictly confidential. We appreciate her dedication and interest in learning.

Velmurugan
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